Lesson notes · for you

How machines learn

AI · Level 1 · Lesson 2

Time	Materials
25 min	Paper and a pen; A friend or family member to play with; Squared paper, if you have some

Learning goals

- Put the five steps of machine learning in order.
- Explain why a model is tested on examples it has never seen.
- Explain why more, and more varied, examples help.

Activity: Be the machine

1. Choose (3 min): Pick a secret pattern that is easy to draw, such as “has a dot inside”, “has exactly three corners” or “is shaded”. Write it down and keep it hidden.
2. Train (6 min): Draw 8 shapes that follow the pattern under the label “yes” and 8 that don’t under “no”. Vary size, direction and shading in both groups, so only the pattern tells them apart. Your partner studies them for 2 minutes: this is the training data.
3. Test (6 min): Hide the training drawings. Draw 6 new shapes one by one, including one or two tricky ones. Your partner says “yes” or “no” for each: this is the test on examples they have never seen.
4. Score (4 min): Count the right answers out of 6, then reveal the pattern. Which shape fooled them, and which extra training example would have helped?
5. Swap and talk (6 min): Swap roles and play again. Then name the five steps you used: collect, label, train, test, use. Say why testing with the same 16 drawings would have been too easy.

You need paper and a pen for each pair. Plan the secret pattern and the 6 test shapes before you start, and make sure at least one test shape looks different from all the training shapes.

Discussion

- What would you teach a machine to sort, and what labels would you use?
- Why is memorising last year’s exam paper a risky way to revise?
- Whose messages, voices or photos might be missing from examples collected only at your school?

Check yourself

- Write the five steps in order, and one reason to test on new examples.

Try it this week

Play “Be the machine” with a friend or someone at home: draw shapes that follow a secret pattern and shapes that don’t, let them find the pattern, then test them with new shapes.

Worksheet

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Name _____ Class _____ Date _____

1 **A scam filter learned only from messages promising free data. A fake scholarship offer arrives. What's most likely?**

◯ It learns the new trick by itself straight away ◯ It might miss it ◯ It catches it easily

2 **Put the five steps in order.**

☐ Test on examples it never saw

☐ Use it on new messages

☐ Collect examples

☐ Train: find the patterns

☐ Label each one

3 **Your filter scores 18 out of 20 on the messages you kept aside. That tells you more than 20 out of 20 on its training messages would.**

◯ True ◯ False

4 **Complete the sentence.**

We _____ each example with the right answer, _____ the model to find patterns, then _____ it on examples it has never seen.

Word bank: memorise · test · label · delete · train

✎ **What would you like to teach a machine to sort, at school or at home? Which examples and labels would it need?**

Check your answers

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1 **It might miss it**

It only knows the kinds of scams it was shown, and a scholarship scam looks different from the free-data ones.

2 **Top to bottom: 4, 5, 1, 3, 2**

You can't label what you haven't collected, or train without labels.

3 **True**

The kept-aside messages are new to it, like a real exam, so 18 out of 20 shows it learned patterns that work on messages it has never seen.

4 **We label each example with the right answer, train the model to find patterns, then test it on examples it has never seen.**

Label, train, test. The test uses examples the model never saw, to check that it learned the patterns instead of memorising its examples.